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In [1]: # Define a sample string
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In [2]: text = "PythonProg"
```

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In [3]: print("first character of text(Positive index 5):", text[5])
```

first character of text(Positive index 5): n

```
In [4]: print("last character(Negative index -1):",text[-1])  
print("last character(Negative index -4):",text[-4])
```

last character(Negative index -1): g
last character(Negative index -4): P

```
In [5]: print("first character of text(Positive index 0):", text[0])
```

first character of text(Positive index 0): P

```
In [6]: print("__ string Slicing __")
```

_ _ string Slicing _ _

```
In [7]: print("Slice[0:6]:",text[0:6])
```

Slice[0:6]: Python

```
In [8]: print("Slice[:6]:",text[:6])
```

Slice[:6]: Python

```
In [9]: print("Slice[6:]:",text[6:])
```

Slice[6:]: Prog

```
In [10]: print("Slice[-4:-1]:",text[-4:-1])
```

Slice[-4:-1]: Pro

```
In [11]: print("Slice[::2]:",text[::2])
```

Slice[::2]: PtoPo

```
In [12]: print("Reversed string[::-1]:",text[::-1])
```

Reversed string[::-1]: gorPnohtyP

```
In [13]: # Define a sample messy string
```

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In [14]: phrase = "Learning Python is fun!"
```

```
In [15]: print("Original Phrase:", repr(phrase))
```

Original Phrase: 'Learning Python is fun!'

```
In [16]: print("1.Upper case:",phrase.upper())
```

1.Upper case: LEARNING PYTHON IS FUN!

```
In [17]: print("1.Lower case:",phrase.lower())
```

1.Lower case: learning python is fun!

```
In [18]: cleaned_phrase = phrase.strip()
```

```
In [19]: print("2. Cleaned (strip):", repr(cleaned_phrase))
```

2. Cleaned (strip): 'Learning Python is fun!'

```
In [20]: replaced_text = cleaned_phrase.replace("fun", "awesome")
```

```
In [21]: print("3. Replaced Text:", replaced_text)
```

3. Replaced Text: Learning Python is awesome!

```
In [22]: word_list = cleaned_phrase.split(" ")
```

```
In [23]: print("4. Split into List:", word_list)
```

4. Split into List: ['Learning', 'Python', 'is', 'fun!']

```
In [24]: joined_text = "-".join(word_list)
```

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In [25]: print("5. Joined with Hyphen:", joined_text)
```

5. Joined with Hyphen: Learning-Python-is-fun!

```
In [26]: print("6. Index of 'Python' (find):", cleaned_phrase.find("Python"))
```

6. Index of 'Python' (find): 9

```
In [27]: print("7. Count of letter 'n':", cleaned_phrase.count("n"))
```

7. Count of letter 'n': 4

```
In [28]: print("8. Starts with 'Learning'?:", cleaned_phrase.startswith("Learning"))  
         print(" Ends with '!'?:", cleaned_phrase.endswith("!"))
```

```
8. Starts with 'Learning'? : True  
  Ends with '!'?: True
```

```
In [29]: print("9. Is '12345' numeric?:", "12345".isdigit())  
         print(" Is 'Python' purely alphabetic?:", "Python".isalpha())
```

```
9. Is '12345' numeric?: True  
  Is 'Python' purely alphabetic?: True
```

```
In [ ]:
```